

GOSFORD HIGH SCHOOL.
Extension 2 Mathematics.
Assessment task December 2005.

Question 1.

(a) Given the complex numbers $A = 3 - 4i$ and $B = 1 + i$,
determine the following in the form $x + iy$.

- | | |
|--------------------|---|
| (i) $A - B$ | 1 |
| (ii) $\frac{A}{B}$ | 2 |
| (iii) B^2 | 2 |
| (iv) $\sqrt{A}$ | 3 |

(b) Given $C = 1 + \sqrt{3}i$

- | | |
|--|---|
| (i) Write C in mod-arg form. | 2 |
| (ii) Hence, using De Moivre's theorem find C^6 . | 2 |

(c) Factorise $z^2 + 4z + 5$ over the complex field. 2

(d) Given that $1, \omega, \omega^2$ are the cubed roots of unity show that:
 $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega) = 3$ 2

(e) Solve the equation $z^2 + 2\bar{z} + 6 = 0$ giving the solutions in the
form $z = x + iy$ where x and y are real numbers 3

(f) Draw a neat accurate sketch of the following.

(i) $\operatorname{Re}(z) = 4$ 1

(ii) $z\bar{z} - 4(z + \bar{z}) = 10$ 3

(iii) $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{2}$ 2

Question 2.

(a) By using implicit differentiation show that the equation of the tangent to the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (a and b constants) at the point $(a \cos \theta, b \sin \theta)$ is given by $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 3

(b) Find the gradient of the curve $2x^3 - x^2y + y^3 = 1$ at the point $(2, -3)$. 3

(c) If α, β, γ are the roots of $x^3 + px + q = 0$, find in terms of p and q :

(i) $\alpha + \beta + \gamma, \quad \alpha\beta + \beta\gamma + \alpha\gamma, \quad \alpha\beta\gamma$ 1

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

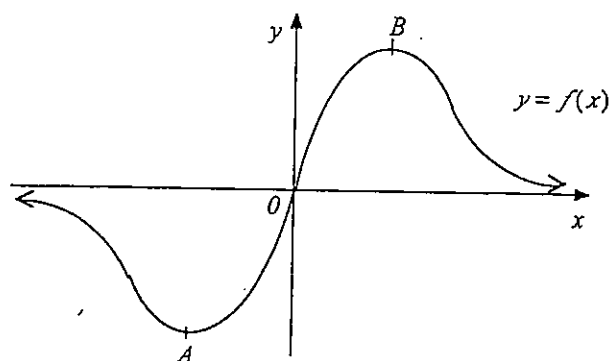
(iii) $\alpha^3 + \beta^3 + \gamma^3$ 2

(d) Given that $-1 + i\sqrt{3}$ is a root of $3x^3 + 5x^2 + 10x - 4 = 0$ find the other roots. 2

Question 3.

(a) the diagram below shows the graph of the function

$$f(x) = xe^{-\frac{1}{2}x^2}.$$



(i) Find the coordinates of the stationary points A and B . 3

(ii) Find the gradient of the tangent to the curve at the origin O .
Hence find the set of values of the real number k such that the equation $f(x) = kx$ has three real roots. 2

(iii) On separate diagrams sketch the following graphs.

(α) $y = |f(x)|$ 2

(β) $y = \{f(x)\}^2$ 2

(γ) $y = \frac{1}{f(x)}$ 2

(χ) $y = \sqrt{f(x)}$ 2

(λ) $y^2 = f(x)$ 2

(θ) $y = f'(x)$ 2

(b) Sketch the graph of $y = \frac{1}{1 + e^{\frac{1}{x}}}$ 3

(c) Solve graphically $2\cos|x| > 1$ for $-\pi \leq x \leq 2\pi$. 3

Question 4.

(a) (i) Sketch the intersection of the locus described by $|z| \leq 3$ and $-\frac{\pi}{4} \leq \arg(z+3) \leq \frac{\pi}{4}$. 3

(ii) If the complex number ω lies on the boundary of the region sketch in part (i), find the minimum value of $|\omega|$. 2

(b) OABC is a rectangle on the argand diagram in which side OC is twice the length of OA, where O is the origin.

(i) If A represents the complex number $1+2i$, find the complex numbers represented by B and C given that the argument of the complex number represented by the point C is negative. 2

(ii) If the rectangle is rotated anticlockwise $\frac{\pi}{3}$ radians about O, find the complex number represented by the new position of A. 1

(c) Solve $z^4 + 16 = 0$ over the complex numbers giving your answer in $x + iy$ form. 2

(d) If Z_1, Z_2, Z_3, Z_4 and Z_5 are the roots of $Z^5 = 1$

(i) Determine the values of all these roots in the form $\cos\theta + i\sin\theta$. 2

(ii) Factorise $Z^5 = 1$ in terms of quadratic and linear factors. 2

(iii) Hence show that $\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$ 2

(e) Given $z = \cos \theta + i \sin \theta$

i) Show $z^n + \frac{1}{z^n} = 2 \cos n\theta$. 2

ii) Hence by expanding $(z + \frac{1}{z})^4$ find an expression
for $\cos^4 \theta$ in the form $a \cos 4\theta + b \cos 2\theta$. 3

COMPLEX NUMBERS

1) If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

Then: $\bar{z}_1 =$

$$|z_1| =$$

$$\arg(z) =$$

$$|z_1 z_2| =$$

$$\arg(z_1 z_2) =$$

$$\left| \frac{z_1}{z_2} \right| =$$

$$\arg\left(\frac{z_1}{z_2}\right) =$$

2) If $z = r(\cos \theta + i \sin \theta)$

then $z^n =$

3) If ω is a complex cube root of unity then:

a) $\omega^3 =$

b) $1 + \omega + \omega^2 =$

POLYNOMIALS

Remainder theorem.

If $P(x)$ is divided by $(x - a)$ then the remainder is

Factor theorem

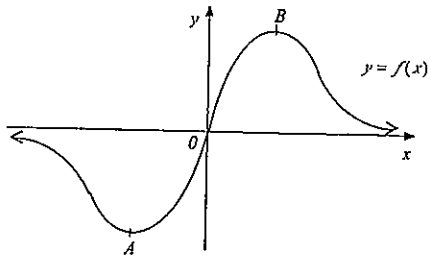
If $(x - a)$ is a factor of $P(x)$ then

Roots of multiplicity

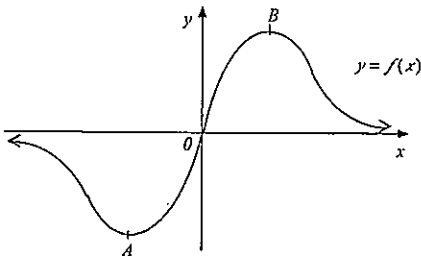
If $x = a$ is an n fold root of $P(x)$ then $x = a$ is an fold root of

If a complex number $a + ib$ is a root of a polynomial, whose coefficients are then is also a root.

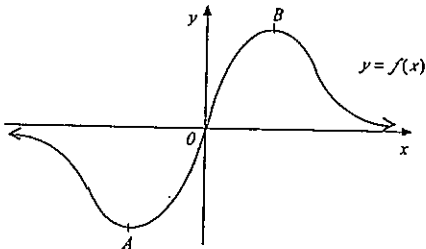
(α) $y = |f(x)|$



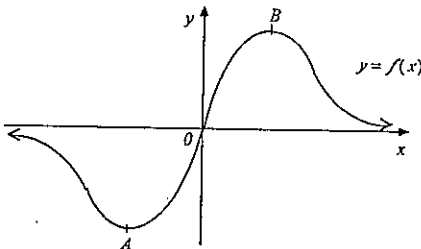
(χ) $y = \sqrt{f(x)}$



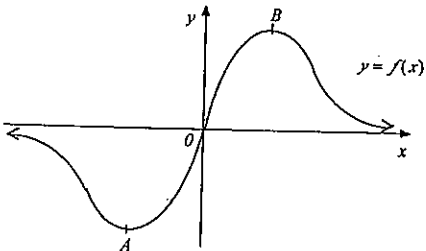
(β) $y = \{f(x)\}^2$



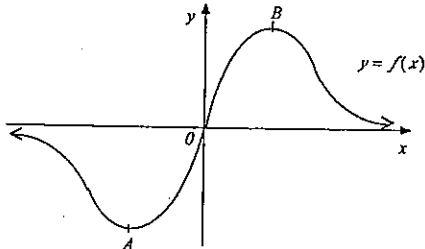
(λ) $y^2 = f(x)$



(γ) $y = \frac{1}{f(x)}$



(θ) $y = f'(x)$



$$1a) (3-4i) - (1+i) = 2-5i$$

$$ii) \frac{3-4i}{1+i} \times \frac{1-i}{1-i} = \frac{3-4- (3+4)i}{2} = -\frac{1}{2} + \frac{7}{2}i$$

$$iii) (1+i)^2 = 1+2i+2i = 2i$$

$$iv) \sqrt{3-4i} = x+iy$$

$$3-4i = x^2 - y^2 + 2xyi$$

$$x^2 - y^2 = 3$$

$$xy = -2 \Rightarrow y = -\frac{2}{x}$$

$$x^2 - \frac{4}{x^2} = 3 \Rightarrow x^4 - 3x^2 - 4 = 0$$

$$(x^2-4)(x^2+1) = 0$$

$$x = \pm 2 \text{ or } \pm i \text{ (not real)}$$

$$y = \mp 1$$

$$\therefore \sqrt{3-4i} = \pm 2 \mp i$$

$$b) i) 1+\sqrt{3}i = 2 \operatorname{cis} \frac{\pi}{3}$$

$$ii) \left(2 \operatorname{cis} \frac{\pi}{3}\right)^6 = 64 \operatorname{cis} 2\pi = 64$$

$$c) z^2 + 4z + 4 + 1 = 0$$

$$(z+2)^2 + 1 = (z+2)^2 - i^2$$

$$(z+2+i)(z+2-i)$$

$$d) z^3 = 1 \Rightarrow z^3 - 1 = 0$$

$$(z-1)(z^2+z+1) = 0$$

$$z=1, \text{ or } \omega \text{ or } \omega^2$$

$$\omega \neq 1, \therefore \omega^2 + \omega + 1 = 0$$

$$(1+2\omega+3\omega^2) = (1+\omega+\omega^2+\omega+2\omega^2)$$

$$= \omega + 2\omega^2 = \omega + \omega^2 + \omega^2$$

$$= \omega^2 - 1$$

$$(1+2\omega^2+3\omega) = 1+\omega+\omega+\omega+2\omega$$

$$= \omega^2 + \omega + \omega = \omega - 1$$

$$(\omega^2-1)(\omega-1) = (\omega-1)(\omega+1)(\omega-1)$$

$$= \omega^3 - \omega^2 - \omega + 1$$

$$= 1 - (\omega^2 + \omega + 1) + 1 + 1 = 3$$

$$e) (x+iy)^2 + 2(x-iy) + 6 = 0$$

$$x^2 - y^2 + 2x + 6 = 0$$

$$2xyi - 2iy = 0$$

$$2yi(x-1) = 0$$

$$x=1 \text{ or } y=0$$

$$\text{When } x=1, y^2 = 1+2+6$$

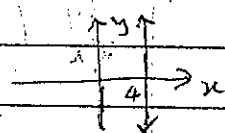
$$y = \pm 3$$

$$\text{When } y=0, x^2 + 2x + 6 = 0$$

No solutions (real)

$$z = 1+3i \text{ or } 1-3i$$

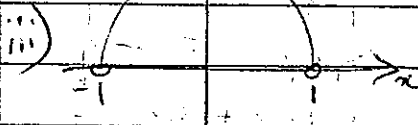
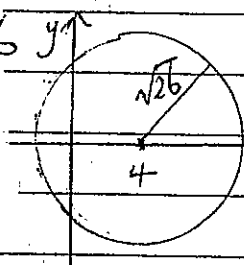
$$f) i) x=4$$



$$ii) x+y-4(2x)=10$$

$$x^2 - 8x + 16 + y^2 = 26$$

$$(x-4)^2 + y^2 = 26$$



$$x^2 + y^2 = 1 \text{ with } y > 0$$

$$2) \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

$$\text{at } (a \cos \theta, b \sin \theta) =$$

$$\frac{dy}{dx} = \frac{-b^2 \cos \theta}{a^2 \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta y - a b \sin^2 \theta = -b \cos \theta x + a b \cos^2 \theta$$

$$b \cos \theta x + a \sin \theta y = a b (\sin^2 \theta + \cos^2 \theta)$$

$$= a b$$

$$\div b \text{ by } a b$$

$$x \cos \theta + y \sin \theta = 1$$

$$b) \quad 2x^3 - x^2 y + y^3 = 1$$

$$6x^2 - x^2 \frac{dy}{dx} - y \cdot 2x + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (3y^2 - x^2) = 2xy - 6x^2$$

$$\frac{dy}{dx} = \frac{2xy - 6x^2}{3y^2 - x^2}$$

$$\text{At } (2, -3), \frac{dy}{dx} = \frac{-12 - 24}{27 - 4}$$

$$= \frac{-36}{23}$$

$$c) \quad i) \quad \alpha + \beta + \gamma = 0$$

$$\alpha \beta + \beta \gamma + \alpha \gamma = p$$

$$\alpha \beta \gamma = -q$$

$$ii) \quad \frac{\beta \gamma + \alpha \gamma + \alpha \beta}{\alpha \beta \gamma} = -\frac{p}{q}$$

$$iii) \quad \alpha \text{ is a root}$$

$$\alpha^3 + p\alpha + q = 0 \quad \text{--- (1)}$$

$$\beta^3 + p\beta + q = 0 \quad \text{--- (2)}$$

$$\gamma^3 + p\gamma + q = 0 \quad \text{--- (3)}$$

$$\alpha^3 + \beta^3 + \gamma^3 + p(\alpha + \beta + \gamma) + 3q = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = 0 \quad \text{--- } 3q = 3q$$

$$-1 + i\sqrt{3}, -1 - i\sqrt{3} \text{ (coeffs are real)}$$

$$\text{Sum of roots} = -2 + \gamma = -\frac{5}{3}$$

$$\gamma = 2 - \frac{5}{3} = \frac{1}{3}$$

$$\text{Roots are } -1 + i\sqrt{3}, -1 - i\sqrt{3}, \frac{1}{3}$$

$$f(x) = x e^{-tx^2} - x + e^{-tx^2}$$

$$= e^{-tx^2} (-x^2 + 1)$$

$$f'(x) = 0 \text{ for start pts}$$

$$x = \pm 1 \quad y = \pm e^{-\frac{1}{2}} = \pm \frac{1}{e}$$

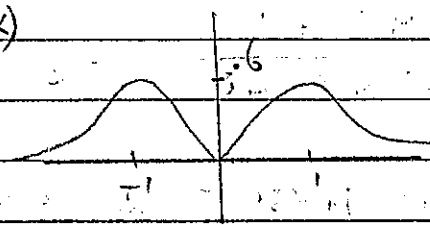
$$A(-1, -\frac{1}{e}), B(1, \frac{1}{e})$$

$$ii) \quad \text{When } x=0, f'(0) = e^0 = 1$$

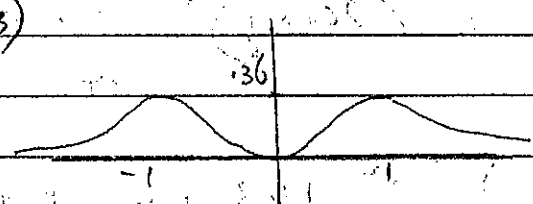
$$y = kx \text{ intersects 3 times}$$

$$0 < k < 1$$

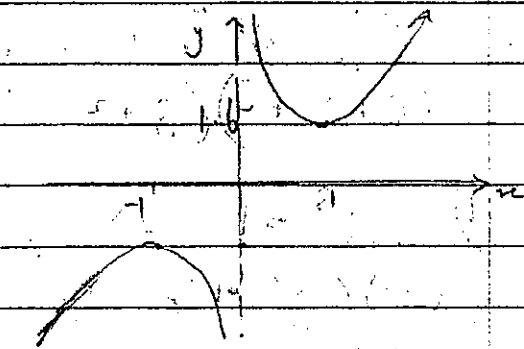
$$iii) \quad \alpha)$$



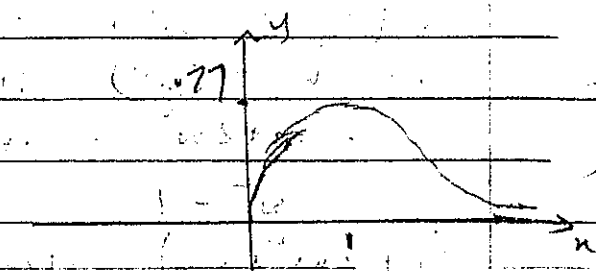
$$\beta)$$



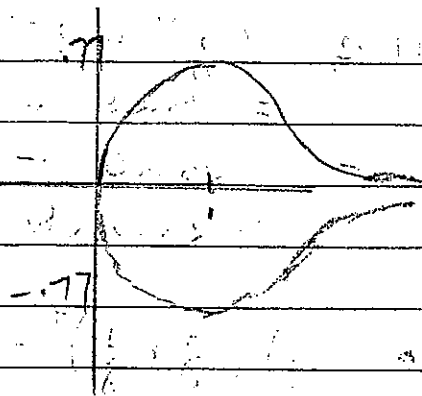
$$\gamma)$$



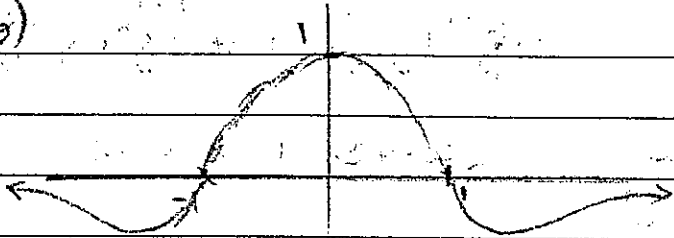
$$\delta)$$



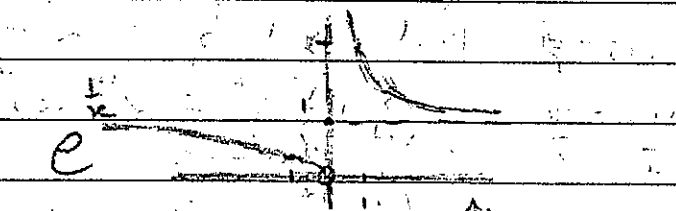
2)



3)

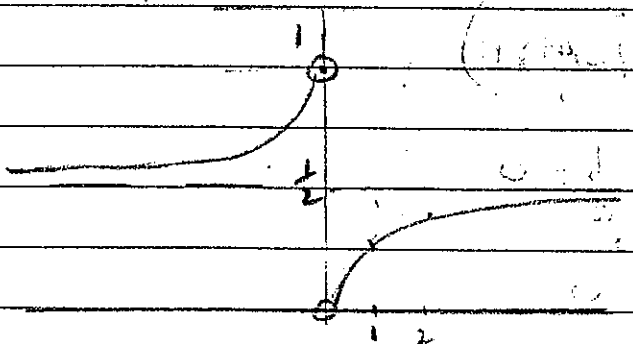


b)



$$1 + e^{-1/x}$$

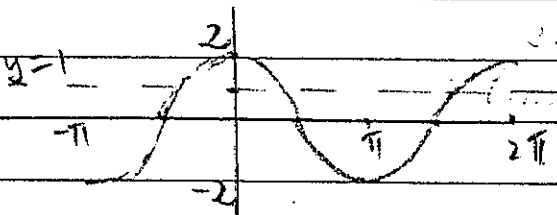
$$y = \frac{1}{1 + e^{-1/x}}$$



$$y' = -\left(1 + e^{-1/x}\right)^{-2} \cdot e^{-1/x} \cdot \frac{1}{x^2} = -\frac{e^{-1/x}}{x^2 (1 + e^{-1/x})^2}$$

Infinite gradient at $x=0$

c)

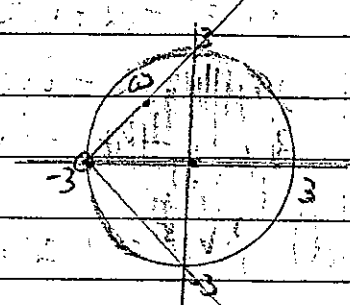


$$\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$$

$$\cos |x| > \frac{1}{2}$$

$$-\frac{\pi}{3} < x < \frac{\pi}{3} \quad \text{or} \quad \frac{5\pi}{3} < x < 2\pi$$

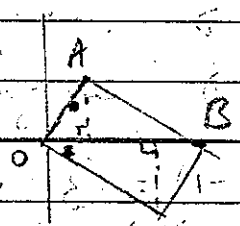
4 a)



i) Min Val (w) is perp dist from 0 to $y = x + 3$

$$d = \frac{|0 - 0 + 3|}{\sqrt{1^2 + 1^2}} = \frac{3}{\sqrt{2}}$$

b)



$$|OC| = 2|OA|$$

Transform A $z \rightarrow 2z$ + rotate $-\frac{\pi}{2}$ i.e multiply by $-i$

$$z_3 = (-i)(2)(1+2i) = 4-2i$$

$$z_2 = (1+2i) + (4-2i) = 5$$

iii) $|z| = \sqrt{5}$ and $\arg z = \frac{\pi}{4}$ is equilateral

$$\text{multiply A by } \text{cis } \frac{\pi}{3} \\ (1+2i) \left(\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) = \frac{1}{2} - \sqrt{3} + i \left(\frac{\sqrt{3}}{2} + 1 \right)$$

d)

$$z^4 = -16 = 16 \text{ cis } \pi$$

$$z = 2 \text{ cis } \frac{\pi + 2k\pi}{4}$$

$$z = 2 \text{ cis } \frac{\pi}{4}, 2 \text{ cis } \frac{3\pi}{4}, 2 \text{ cis } \frac{5\pi}{4}, 2 \text{ cis } \frac{7\pi}{4}$$

c) Cont. $z = \frac{2}{\sqrt{2}} + i \frac{2}{\sqrt{2}} = \sqrt{2} + i\sqrt{2}$

$z_2 = -\sqrt{2} + i\sqrt{2}$

$z_3 = \sqrt{2} - i\sqrt{2}$

$z_4 = -\sqrt{2} - i\sqrt{2}$

$\therefore z^n + z^{-n}$

$= \cos n\theta + i \sin n\theta +$

$\cos n\theta - i \sin n\theta$

$= 2 \cos n\theta$

d) i) $z^5 = 1 \text{ cis } 0$

$z = 1 \text{ cis } \frac{2k\pi}{5}$

$k=0 \quad z = 1$

$k=1 \quad z = \text{cis } \frac{2\pi}{5} = \omega$

$k=2 \quad z = \text{cis } \frac{4\pi}{5} = \omega^2$

$k=3 \quad z = \text{cis } \frac{6\pi}{5} = \text{cis } -\frac{4\pi}{5}$

$k=4 \quad z = \text{cis } \frac{8\pi}{5} = \text{cis } -\frac{2\pi}{5} = \bar{\omega}$

ii) $(z^5 - 1) = (z - 1)$

$(z - \omega)(z - \bar{\omega})(z - \omega^2)(z - \bar{\omega}^2)$
 $(z - 1) \left(z^2 - (\omega + \bar{\omega})z + \omega\bar{\omega} \right)$
 $(z^2 - (\omega^2 + \bar{\omega}^2)z + \omega^2\bar{\omega}^2)$

$= (z - 1) \left(z^2 - 2\cos\frac{2\pi}{5}z + 1 \right) \left(z^2 - 2\cos\frac{4\pi}{5}z + 1 \right)$

iii) Sum of roots $= -\frac{b}{a} = 0$

$1 + \omega + \bar{\omega} + \omega^2 + \bar{\omega}^2 = 0$

$+ 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$

$\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$

e) $z = \cos\theta + i \sin\theta$

$z^{-1} = (\cos\theta + i \sin\theta)^{-1}$
 $= \cos -\theta + i \sin -\theta$

$z^n = \cos n\theta + i \sin n\theta$

$z^{-n} = \cos -n\theta + i \sin -n\theta$

(by De Moivre's Theorem)

But $\cos -n\theta = \cos n\theta$

$\sin -n\theta = -\sin n\theta$

ii) $\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^3z^{-1} + 6z^2z^{-2} + 4z^1z^{-3} + z^{-4}$
 $= (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6$

$= 2\cos 4\theta + 8\cos 2\theta + 6$

But $z + \frac{1}{z} = 2\cos\theta$

$\therefore \left(z + \frac{1}{z}\right)^4 = 2^4 \cos^4\theta$

$\therefore 16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$

$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$